

Healthcare Analytics Platform - Project Overview

Project Objective

Build an end-to-end Healthcare Analytics Platform using Databricks Free Edition implementing Medallion Architecture, Delta Lake, Incremental ETL, Streaming, Audit Logging, Error Handling, Governance and Analytics.

Technology Stack

- Databricks Free Edition
- PySpark
- Delta Lake
- SQL
- GitHub
- Unity Catalog
- Databricks Workflows
- Structured Streaming

Datasets

- Patients
- Claims
- EMR
- Pharmacy
- Appointments
- Bed Management
- ICU Vital Stream
- Hospitals
- Doctors
- Departments
- Diagnosis Codes
- Procedure Codes
- Insurance Providers

Architecture Flow

Source Files → Validation → Quarantine → Bronze → Silver → Incremental Processing (MERGE/CDC/SCD) → Gold → Dashboards

Project Structure

source_files
quarantine
bronze
silver

gold
checkpoints
logs
archive

Notebook Flow

1. 01_setup
2. 02_bronze_ingestion
3. 03_validation_quarantine
4. 04_silver_transformation
5. 05_incremental_merge_cdc
6. 06_gold_analytics
7. 07_streaming_pipeline
8. 08_dashboard_queries

Key Features

- Medallion Architecture
- Incremental ETL
- MERGE / CDC
- Schema Validation
- Business Rule Validation
- Quarantine
- Audit Logs
- Error Logs
- Pipeline Logs
- Streaming with Checkpoints
- Delta Lake (Time Travel, OPTIMIZE, VACUUM)
- Performance Optimization
- GitHub Integration

Gold Outputs

- Patient Visit Analytics
- Claims Analytics
- EMR Analytics
- Pharmacy Analytics
- Appointment Analytics
- Bed Occupancy Analytics
- ICU Alerts
- Hospital KPI Dashboard

Optimizations

- Partitioning
- Repartitioning

- Caching
- Incremental MERGE
- Checkpointing
- Delta Optimization
- Quarantine Framework
- Audit Logging

Conclusion

This project demonstrates an enterprise-ready healthcare data engineering solution covering ingestion, transformation, incremental processing, streaming, governance, monitoring and reporting.